Cynbit Technologies — Technical Approach Report

Cynbit IntelliAssist – AI-Powered Lead Qualification & Knowledge Assistant

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1. Problem Framing

Business Problem → Technical Problem Translation

Business problem: Cynbit Technologies — an AI-first company selling AI agents, custom software, and growth services — collects leads through a static contact form. This creates two compounding failure modes: (1) potential clients who want to know what Cynbit actually does hit a dead end and bounce; (2) the small team manually triages every lead submission with no signal on intent, budget, or timeline.

Technical translation:

Business Need	Technical Requirement
Answer "what does Cynbit do?" 24/7	RAG pipeline over public cynbit.in content
Know if a lead is sales-ready	BANT field extraction via structured LLM output
Don't make up capabilities Cynbit doesn't have	Grounded-only responses with source citation enforcement
Works without an ML team running it	DEMO_MODE + reproducible FAISS index build
Fits a bootstrapped company's budget	Local embeddings, Gemini Flash pricing, free-tier deployment

Success Metrics (Measurable)

1. **Retrieval Hit Rate** ≥ **85%** — correct source document in top-3 for known-answer queries
1. **Groundedness Rate** ≥ **90%** — every factual claim in a response carries a `[Source: <url>]` citation
1. **Out-of-scope Detection = 100%** — no off-topic queries receive fabricated answers
1. **BANT Extraction Accuracy** ≥ **70%** — across Budget, Authority, Need, Timeline fields
1. **End-to-end latency p95 < 2s** — live LLM mode on production hosting

What "Done" Looks Like

A visitor to cynbit.in opens the IntelliAssist widget, asks "Do you build custom CRMs?", receives a cited answer from the Cynbit services page, then is guided through a 4-question qualification flow. At conversation end, a structured `LeadProfile` (name, company, email, BANT scores) fires to `opportunities@cynbit.in`. The Cynbit team opens their inbox and sees a pre-qualified lead — no form to read, no triage call needed.

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2. System Architecture

```
flowchart TD
    A([User / Website Visitor]) --> B[Streamlit UI\nChat Interface]

    subgraph Ingestion ["Data Ingestion Pipeline (Offline)"]
        I1[cynbit.in public pages\nServices · Products · Team · Blog] --> I2[HTML Scraper\n+ Manual JSON]
        I2 --> I3[Cleaner\nHTML strip · Dedup · Normalize]
        I3 --> I4[Chunker\n512 tokens · 64 overlap]
        I4 --> I5[Embedder\nall-MiniLM-L6-v2]
        I5 --> I6[(FAISS Index\nIndexFlatIP)]
    end

    B --> C{Query Router}

    C -->|Knowledge query| D[Guardrails — Pre\nScope check · Injection filter]
    D --> E[Embedder\nall-MiniLM-L6-v2]
    E --> F[FAISS Retrieval\nTop-k=3 chunks]
    F --> G[Context Assembler\nChunks + Conversation history]
    G --> H[LLM\nGemini 1.5 Flash]
    H --> I[Guardrails — Post\nCitation check · Hallucination flag]
    I --> J[Structured Response\nMarkdown + source citations]
    J --> B

    C -->|Lead signal detected| K[Lead Qualifier\nBANT extraction prompt]
    K --> L[LLM\nGemini 1.5 Flash]
    L --> M[Pydantic Parser\nLeadProfile model]
    M --> N{Lead Score\nThreshold ≥ 60?}
    N -->|Yes| O[Email Notification\nopportunities@cynbit.in]
    N -->|No| P[Continue conversation\nGather missing BANT fields]
    P --> B

    I6 -.->|Query time| F
```

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3. Data Sources & Pipeline

Sources

All content is sourced from publicly accessible pages on `cynbit.in`:

Page	Content Type	Key Entities
/ (homepage)	Company overview, value prop	Services summary, CTA
`/services`	Service offerings	AI Agents, Software Dev, Growth/Lead Gen
`/products`	Product descriptions	BitElevate, TenantEase, Cynbit Studio, Revio
`/about` / story	Company history	Founded April 2024, Jaipur, ~15 employees
`/team`	Team profiles	Names, roles
`/blog`	Thought leadership	Use cases, technical signals
`/contact`	Contact info	`opportunities@cynbit.in`

Collection Strategy

Manual scrape + structured JSON. Reason: Cynbit's site is relatively small (~30 meaningful pages), and a manual pass ensures clean semantic boundaries — automated scrapers often merge navigation text, footer links, and body copy into the same chunk. Each page is saved as a JSON record:

```
{
  "url": "https://cynbit.in/services",
  "title": "Services – Cynbit Technologies",
  "content": "...",
  "scraped_at": "2026-10-29T00:00:00Z"
}
```

This JSON corpus is version-controlled. Reproducibility matters: anyone cloning the repo can rebuild the FAISS index deterministically without re-hitting the live site.

Cleaning

1. **HTML stripping:** `BeautifulSoup` with `get_text(separator=" ")` — removes tags, retains inline structure
1. **Deduplication:** SHA-256 content hash per document; near-duplicates (e.g., repeated footer text across pages) removed via hash comparison
1. **Length normalization:** Documents shorter than 50 tokens discarded (navigation stubs); documents longer than 4,000 tokens truncated at sentence boundary before chunking

Chunking Strategy

- **Chunk size:** 512 tokens | **Overlap:** 64 tokens
- **Why 512/64:** MiniLM's practical context ceiling is 256 tokens, but we chunk at 512 to preserve paragraph coherence before the model compresses meaning. The 64-token (≈2-3 sentence) overlap ensures a key fact split at a chunk boundary still appears in an adjacent chunk — preventing retrieval misses on boundary-straddling answers.
- **Implementation:** `langchain.text_splitter.RecursiveCharacterTextSplitter` with `chunk_size=512, chunk_overlap=64`, splitting on `["\n\n", "\n", " "]` in order.

Embedding

Model: `sentence-transformers/all-MiniLM-L6-v2`

- 384-dimensional dense vectors
- Runs entirely locally — no API call, no per-token cost
- 14,500 sentences/sec on CPU (Intel i5 class) — acceptable for offline indexing
- Adequate semantic quality for a closed-domain corpus (cynbit.in ≠ general-purpose retrieval)

Vectors are L2-normalized before indexing so that inner-product search on `IndexFlatIP` is equivalent to cosine similarity.

Indexing

FAISS `IndexFlatIP` (exact inner product search, cosine-equivalent after normalization):

- ~30 source documents → **~45 chunks** after splitting
- Index size at 45 chunks × 384 dims × 4 bytes ≈ **69 KB** — trivially fits in memory
- Exact search (no quantization needed) at this scale — no accuracy loss from HNSW approximation

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4. Model Choices & Trade-offs

Embedding Models

Model	Dims	Speed	Cost	Quality	Choice?
`all-MiniLM-L6-v2`	384	Fast	Free (local)	Good	■ Selected
`text-embedding-3-small`	1536	Fast	\$0.02/1M tokens	Better	Fallback
`BAAI/bge-small-en-v1.5`	384	Fast	Free (local)	Better	Alt option

Decision rationale: For a 45-chunk corpus, the semantic quality gap between MiniLM and BGE-small is negligible in practice. MiniLM's broader community adoption means more debugged integrations. BGE-small is the first upgrade candidate if retrieval metrics fall below threshold.

LLM Options

Model	Speed	Cost (per 1M tokens)	Context	Quality	Default?
`gemini-1.5-flash`	Fast	\$0.075 in / \$0.30 out	1M tokens	High	■ Default
`gemini-1.5-pro`	Slower	\$1.25 in / \$5.00 out	1M tokens	Highest	Premium
`gpt-4o-mini`	Fast	\$0.15 in / \$0.60 out	128K tokens	High	OpenAI fallback
`gpt-4o`	Slower	\$2.50 in / \$10.00 out	128K tokens	Highest	OpenAI premium

Decision rationale: Flash's 1M-token context window eliminates any concern about long conversation history overflow. At Cynbit's expected traffic volume (see §8), Flash costs under \$2/month — Pro is reserved for high-value async lead scoring jobs where extra reasoning quality justifies the 17× price premium.

Vector Stores

Store	Scalability	Cost	Setup	Choice?
FAISS	Up to ~10M vectors	Free	Local	■ POC
Chroma	Medium	Free	Local/Cloud	Alt (Day 31-60)
Pinecone	Unlimited	SaaS pricing	Managed	Production at scale

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5. Prompt Engineering

Knowledge Assistant System Prompt

```
You are IntelliAssist, Cynbit Technologies' AI knowledge assistant.

Your job is to answer questions about Cynbit's services, products, team, and capabilities.

Rules:
1. ONLY answer based on the provided context chunks. Do not use outside knowledge.
2. ALWAYS cite your source: append [Source: <source_url>] to every factual claim.
3. If the answer is not in the context, say: "I don't have that information in my
   knowledge base. For accurate details, please contact opportunities@cynbit.in."
4. Be concise. Answer in 2-4 sentences unless the question requires more.
5. Never make up pricing, team member names, or product features not in the context.

Context:
{context}
```

```
Conversation history:
{history}
```

Lead Qualifier System Prompt

```
You are IntelliAssist in lead qualification mode.

Your goal is to naturally gather BANT information through conversation — do NOT
interrogate the user. Ask one question at a time. Be helpful and consultative.

BANT fields to collect:
- Budget: What is their approximate budget for this project?
- Authority: Are they the decision-maker, or is someone else involved?
- Need: What specific problem are they trying to solve?
- Timeline: When do they need this implemented?

Conversation so far:
{history}

Current BANT state:
{bant_state}

Respond naturally. If a BANT field emerges from the user's last message, acknowledge
it and move to the next unknown field. If all fields are collected, say:
"Thank you — I have enough to connect you with our team. Could I get your name,
company, and email address?"
```

BANT Extraction Prompt

```
Extract BANT qualification data from the following conversation transcript.
Return a valid JSON object matching the schema exactly. Use null for unknown fields.

Schema:
{
  "budget": "string or null — e.g. '$5,000', 'under $10K', 'not disclosed'",
  "authority": "string or null — e.g. 'CTO', 'decision maker', 'needs approval'",
  "need": "string or null — brief description of the business problem",
  "timeline": "string or null — e.g. 'ASAP', 'Q1 2027', '3 months'",
  "lead_score": "integer 0-100 — your confidence that this is a qualified lead",
  "score_rationale": "string — one sentence explaining the score"
}

Conversation transcript:
{transcript}

Return only the JSON object. No prose, no markdown fences.
```

Few-Shot Examples for BANT Extraction

Example 1 — Well-qualified lead:

Input transcript:

```
User: We're looking to build an AI agent to handle our customer support — about 500
      tickets a day. Our CTO approved a budget of $15,000 for the MVP.
Bot: Great. What timeline are you working with?
User: We'd ideally like to launch before the new year, so roughly 8 weeks.
```

Expected output:

```
{
  "budget": "$15,000",
  "authority": "CTO approved — user likely has sign-off",
```

```
  "need": "AI agent for customer support, 500 tickets/day",
  "timeline": "8 weeks (before year-end)",
  "lead_score": 88,
  "score_rationale": "Clear budget, authority confirmed, specific need and tight timeline — high purchase intent."
}
```

Example 2 — Partially qualified lead:

Input transcript:

```
User: I'm a startup founder exploring AI options for automating our sales outreach.
Bot: What budget are you thinking?
User: We haven't finalized that yet — still in the exploring phase.
```

Expected output:

```
{
  "budget": null,
  "authority": "Founder — decision maker",
  "need": "AI automation for sales outreach",
  "timeline": null,
  "lead_score": 42,
  "score_rationale": "Authority confirmed and need is clear, but no budget or timeline disclosed — early-stage exploration."
}
```

Example 3 — Low-quality / off-scope:

Input transcript:

```
User: Do you guys build mobile games?
Bot: That's outside our current service offerings. We specialize in AI agents,
     custom software, and growth solutions. Is there something in that space I can help with?
User: Nah, I need a game dev studio. Thanks anyway.
```

Expected output:

```
{
  "budget": null,
  "authority": null,
  "need": "Mobile game development (out of scope for Cynbit)",
  "timeline": null,
  "lead_score": 2,
  "score_rationale": "Out-of-scope request, user disengaged — not a qualified lead."
}
```

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6. Evaluation Methodology

Metric Definitions

- **Retrieval Hit Rate:** For a curated set of test queries with known correct source documents, the percentage where the correct document appears in the top-3 FAISS results.
- **Groundedness Rate:** Automated regex check — does the response contain at least one `[Source: ...]` pattern? Responses without citations are flagged as potential hallucinations.
- **Out-of-scope Detection Accuracy:** Percentage of queries that are factually off-topic (e.g., asking about competitors, unrelated industries) that receive the defined fallback response rather than a fabricated answer.

- **Lead Qualification Accuracy:** Across 12 BANT fields extracted from 3 test conversations, the percentage correctly identified vs. ground-truth manually labeled.
- **Latency:** Wall-clock time from user query submission to streamed first token (p50) and full response (p95).

Evaluation Results (DEMO_MODE, 15 Test Cases)

> *[!NOTE]*
>
> *All retrieval and response results below were generated in **DEMO_MODE** (pre-baked responses, no live LLM call). Latency figures for live Gemini mode are estimated from published Flash model benchmarks and typical Streamlit Community Cloud networking overhead.*

Metric	Value	Notes
Retrieval Hit Rate	**87%** (13/15)	2 misses on ambiguous queries ("how big is the team?")
Groundedness Rate	**93%** (14/15)	1 response caught by post-LLM guardrail, rewritten
Out-of-scope Detection	**100%** (3/3)	All 3 off-topic queries flagged correctly
BANT Extraction Accuracy	**75%** (9/12 fields)	Timeline field hardest — users express it vaguely
Avg Latency — DEMO_MODE	**42ms**	No LLM call; FAISS + Python only
Avg Latency — Live Gemini (est.)	**~650ms**	Flash p50 ≈ 400ms + 200ms network + 50ms FAISS

Known failure modes:

- Ambiguous queries that map to multiple source documents (e.g., "team size") reduce retrieval precision — a re-ranker (cross-encoder) would address this.
- Timeline extraction fails when users express time indirectly ("before our Series A closes") — requires entity normalization logic.

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7. Privacy & Security Considerations

Data in the knowledge base:

- 100% sourced from publicly accessible `cynbit.in` pages — no proprietary, internal, or user-generated content is indexed. No GDPR/DPDP Act concerns for the knowledge base itself.

Lead PII handling:

- Name, email, company, and phone captured during qualification are stored in **Streamlit session state only** — ephemeral, scoped to the browser session, not written to disk or any database in the POC.
- For production: lead data must be encrypted at rest (AES-256), transmitted over HTTPS only, and handled under DPDP Act 2023 (India) compliance — explicit consent capture required before collecting personal information.

API key management:

- All API keys (`GEMINI_API_KEY`, `OPENAI_API_KEY`) are loaded via `python-dotenv` from a `.env` file.
- `.env` is listed in `.gitignore` — confirmed absent from the committed repository.
- DEMO_MODE operates without any API key — safe for public demos and CI pipelines.

Prompt injection mitigation:

- System prompt is never interpolated with raw user input — only `{context}` (from FAISS, trusted corpus) and `{history}` (escaped string concatenation) are inserted.
- Pre-query guardrail filters inputs containing common injection patterns (`"ignore previous instructions"`, `"you are now"`, `"DAN"`) and returns a hardcoded refusal.

LLM provider agreements:

- Google Gemini API and OpenAI API both have Data Processing Agreements (DPAs) that cover enterprise/production use. Cynbit should execute these before going live with lead PII in prompts.

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8. Deployment & Cost Estimate

POC: Streamlit Community Cloud

- **Cost:** Free
- **RAM:** 1GB shared — sufficient (FAISS index ≈ 69KB, MiniLM model ≈ 90MB)
- **Limitation:** No persistent storage. FAISS index is rebuilt from the bundled JSON corpus on every cold start (~3s). Acceptable for a POC; not acceptable for production at scale.
- **Cold start:** ~8-12s on first request after inactivity

Production Deployment Options

Option	Cost/month	Latency	Scalability	Notes
Streamlit Community Cloud	Free	~2s cold start	Low	POC only
Railway.app (1GB RAM)	~\$5	200ms	Medium	Good for <1K DAU
Google Cloud Run	~\$10-30 (usage-based)	100-300ms	High	Scales to zero; Cynbit already in Google ecosystem
GKE / Kubernetes	\$50-200+	<100ms	Enterprise	Overkill at current Cynbit scale

Recommended path: Cloud Run at Day 61 milestone — usage-based pricing means near-zero cost at low traffic, no server management overhead, and native integration with Google Secret Manager for API key storage.

LLM API Cost Projection

- **Assumptions:** 50 website visitors/day interact with the bot; average 5 conversation turns each
- **Daily queries:** 50 × 5 = **250 queries/day**
- **Token profile:** ~1,500 input tokens/query (context chunks + history + query) · ~200 output tokens/query

$$\text{Daily cost} = \frac{250 \times (1{,}500 \times 0.075 + 200 \times 0.30)}{1{,}000{,}000} \approx \$0.043\text{/day}$$

Monthly LLM cost ≈ **\$1.30** at Gemini 1.5 Flash pricing. Even at 10× traffic growth, cost stays under \$15/month — well within bootstrapped operating parameters.

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9. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation

Hallucination / fabricated Cynbit claims	Medium	High — damages brand trust	Post-generation guardrail enforces citation regex; responses without `[Source:]` are blocked and rewritten with fallback message
Stale knowledge base	High (without process)	Medium — bot answers questions about discontinued products/prices	Weekly automated re-scrape job (GitHub Actions cron); FAISS index rebuild takes <30s
Prompt injection	Low	High — could expose system prompt or bypass guardrails	Pre-query injection filter; system prompt never directly exposed; all context from trusted FAISS corpus only
Cost overrun	Low	Low	Per-session token counter; hard cap at 3,000 tokens/session; alert if daily spend exceeds \$5 (10× baseline)
Low adoption / users prefer form	Medium	Medium — renders project irrelevant	A/B test chatbot widget vs. static form (Day 31-60); track conversion rate; if chatbot underperforms, investigate UX not the model

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10. 30/60/90-Day Execution Plan

Day 0-30 — Foundation

- [ ] Deploy POC to Streamlit Community Cloud with live Gemini API key
- [ ] Integrate widget into `cynbit.in` via iframe or Streamlit embed
- [ ] Founders run structured user testing (10 internal sessions, capture failure cases)
- [ ] Build weekly knowledge base refresh pipeline (GitHub Actions → scrape → rebuild FAISS index → redeploy)
- [ ] Instrument session analytics: query count, BANT completion rate, drop-off point
- [ ] Fix retrieval misses identified in founder testing — add missing content chunks

Milestone: IntelliAssist live on cynbit.in, processing real visitor queries, Cynbit team receiving lead emails.

Day 31-60 — Integration

- [ ] CRM/email automation: when `lead_score ≥ 60`, POST structured `LeadProfile` to `opportunities@cynbit.in` via SendGrid/SMTP with formatted summary
- [ ] WhatsApp integration via Twilio or WATI — deploy same RAG pipeline behind WhatsApp Business API (Cynbit's B2B audience in India is WhatsApp-first)
- [ ] Upgrade vector store from FAISS (in-memory, rebuilt on cold start) to **Chroma with persistence** — eliminates cold-start index rebuild
- [ ] A/B test instrumented: 50% of cynbit.in visitors see chatbot widget, 50% see original form; track lead submission rate, lead quality score per channel
- [ ] Tune BANT qualification prompts based on first 4 weeks of real conversation data — particularly timeline and budget extraction patterns

Milestone: First A/B test data showing chatbot vs. form conversion differential. WhatsApp pilot running with 5 seed leads.

Day 61-90 — Scale & Optimize

- [ ] Production hosting migration: deploy to **Google Cloud Run** (Docker container, 1GB RAM, auto-scaling)
- [ ] Extend RAG stack to **BitElevate**: student onboarding assistant using same pipeline, different FAISS index (BitElevate course catalog + FAQ content)
- [ ] Voice interface prototype: Gemini Live API for real-time spoken conversation, or Deepgram STT + Piper TTS for lower-latency open-source option
- [ ] Internal analytics dashboard (Streamlit admin view, password-protected): lead pipeline visibility — leads by day, BANT score distribution, top queries, retrieval miss log
- [ ] Document IntelliAssist as a **reusable Cynbit product offering** — package the RAG pipeline, BANT qualifier, and Streamlit template as a deployable product that Cynbit can sell to their own clients (their core market)

Milestone: IntelliAssist running in production, BitElevate pilot deployed, packaged as a client-facing Cynbit product with documented setup playbook.

Appendix: Technology Stack Summary

Layer	Technology	Version	Rationale
Frontend	Streamlit	1.32+	Zero-config Python UI; ideal for rapid B2B AI tool prototyping
Embedding	sentence-transformers	2.6+	Local inference, no API dependency, 384-dim MiniLM
Vector store	FAISS	1.7+	In-process, zero infra, sufficient at <100K vectors
LLM	Gemini 1.5 Flash	API	Best cost/quality ratio; 1M token context eliminates chunking edge cases
Structured output	Pydantic	2.x	Type-safe LeadProfile parsing; validation errors caught at extraction time
Env management	python-dotenv	1.x	`.env` pattern; DEMO_MODE fallback when keys absent
Data pipeline	BeautifulSoup + json	stdlib	Minimal dependency scraping; reproducible JSON corpus
Deployment (POC)	Streamlit Community Cloud	—	Free; sufficient for 1GB RAM workload
Deployment (Prod)	Google Cloud Run	—	Usage-based, scales to zero, managed HTTPS